



Tose'e Higher Education Institute

Data Science ML AI Course Project

Predicting College Applications Using Machine Learning

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July 2025

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Understanding The Problem

Each year, many students apply to different universities, hoping to get admitted. The number of applications received by these universities is different from one and another. Some of them receive numerous applications, while others don't. In this project, we try to predict the number of applications in different universities, using a dataset that contains various features.

1. What is the main motivation of this project?

The main incentive of this project is to predict the number of applications received by various universities, using a dataset with 18 different features. Several Machine Learning algorithms were used, which will be explained in more detail later.

2. Where can the results be useful?

Predicting the number of applications can be crucial for the educational institutions, specifically because it gives good insights to the managers in charge, and they can make better plans and decisions while they are aware of this figure. Like, for instance, having an idea of how many students are going to apply to your university can help you plan for resources and allocation, and better decisions can be made. Another example, universities can have lots of different financial strategies when they know how many people are wishing to get admitted to their institution. Moreover, not only can the universities benefit from the results, but there are many applicants who can make better plans, provided that they have access to the predictions and the dataset.

3. To whom may the result concern and why?

As we discussed, different educational institutions like universities and people who want to get admitted to these universities can find the results useful. Furthermore, enterprises that provide services for students and universities can find the results valuable.

Data understanding

It is vehemently important to think about data, how they were allocated, and also what do they really mean. This investigation has a significant impact on the results. Thus, it is vital to answer all these questions.

1. Where and how has the data been collected?

This dataset was taken from the StatLib library, which is maintained at Carnegie Mellon University. The dataset was used in the ASA Statistical Graphics Section's 1995 Data Analysis Exposition.

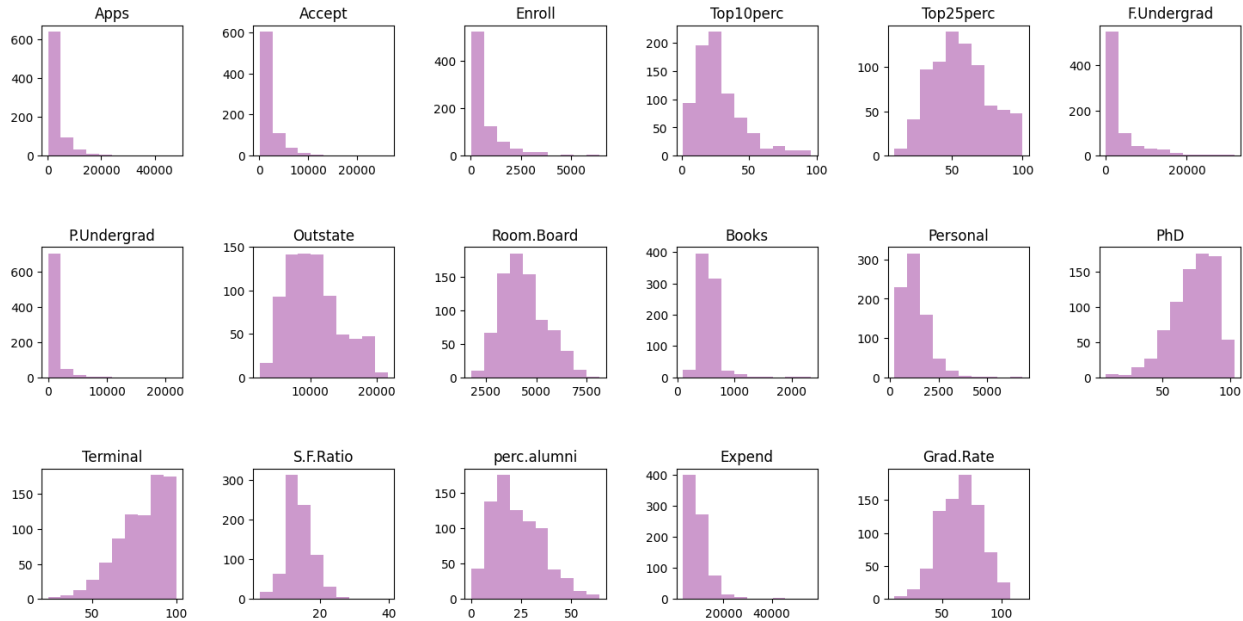
2. What is each variable measuring?

This dataset contains 777 observations and 18 variables. The variable description is as the table below:

Number	Variable	Description of The Variable
1	Private	A factor with levels No and Yes indicating a private or public university
2	Apps	Number of applications received
3	Accept	Number of applications accepted
4	Enroll	Number of new students enrolled
5	Top10perc	Pct. new students from top 10% of H.S. class
6	Top25perc	Pct. new students from top 25% of H.S. class
7	F.Undergrad	Number of full-time undergraduates
8	P.Undergrad	Number of part-time undergraduates
9	Outstate	Out-of-state tuition
10	Room.Board	Room and board costs
11	Books	Estimated book costs
12	Personal	Estimated personal spending
13	PhD	Pct. of faculty with Ph.D.'s
14	Terminal	Pct. of faculty with terminal degree
15	S.F.Ratio	Student/faculty ratio
16	perc.alumni	Pct. alumni who donate
17	Expend	Instructional expenditure per student
18	Grad.Rate	Graduation rate

3. Is there anything ambiguity in the data?

Statistical analyses can investigate that if there is any problem in the dataset or not. These are all the features distribution:



After plotting all the features, we can see that, for instance, there is an observation that has a Grad.Rate more than 100 percent. Since these data were collected by a reliable source, it is justified to assume, there is nothing wrong with the dataset. We assume that the reason why there are observations with a feature of more than 100 percent is because of the definition of the features. Like, for example, in 1995, someone had graduated who had never been admitted to that university initially, hence, he or she had probably transferred to this university. Also, something else which is considerable is two of the universities, both of which are acquiring applications, significantly more than in other observations, however, regarding how large they are, there is no more than usual.

4. Is it possible that any mistakes were made during data collection?

As we discussed, since the source of data is highly reliable, we assume there would be no mistakes in the dataset and data collection process.

5. What other features could contribute to better prediction and understanding

Any other feature or variable that contains some information about the colleges and universities can be helpful. Especially if they have any details about how popular the university is. Typically, more popular universities, whether they have a better rank or are located in a better environment, would have more applicants each year. Therefore, variables such as an index like the global or world ranking could be useful for making a more accurate prediction.

6. What is the type of each variable?

Almost all of the variables were as type of a numeric type. However, only one of the variables was categorical type. Since the scikit-learn library doesn't support categorical variables, we convert them to dummy variables. Here is the table of types and variables:

Number	Variable	Type
1	Private	Categorical
2	Apps	Numeric
3	Accept	Numeric
4	Enroll	Numeric
5	Top10perc	Numeric
6	Top25perc	Numeric
7	F.Undergrad	Numeric
8	P.Undergrad	Numeric
9	Outstate	Numeric
10	Room.Board	Numeric
11	Books	Numeric
12	Personal	Numeric
13	PhD	Numeric
14	Terminal	Numeric
15	S.F.Ratio	Numeric
16	perc.alumni	Numeric
17	Expend	Numeric
18	Grad.Rate	Numeric

7. What is a brief statistical summary of the variables?

The summary of the statistical analysis of the variables is as below:

	max	75%	50%	25%	min	std	mean	count
Apps	48094	3624	1558	776	81	3870.201484	3001.638353	777
Accept	26330	2424	1110	604	72	2451.113971	2018.804376	777
Enroll	6392	902	434	242	35	929.17619	779.972973	777
Top10perc	96	35	23	15	1	17.640364	27.558559	777
Top25perc	100	69	54	41	9	19.804778	55.796654	777
F.Undergrad	31643	4005	1707	992	139	4850.420531	3699.907336	777
P.Undergrad	21836	967	353	95	1	1522.431887	855.298584	777
Outstate	21700	12925	9990	7320	2340	4023.016484	10440.66924	777
Room.Board	8124	5050	4200	3597	1780	1096.696416	4357.526384	777
Books	2340	600	500	470	96	165.10536	549.380952	777
Personal	6800	1700	1200	850	250	677.071454	1340.642214	777
PhD	103	85	75	62	8	16.328155	72.660232	777
Terminal	100	92	82	71	24	14.722359	79.702703	777
S.F.Ratio	39.8	16.5	13.6	11.5	2.5	3.958349	14.089704	777
perc.alumni	64	31	21	13	0	12.391801	22.743887	777
Expend	56233	10830	8377	6751	3186	5221.76844	9660.171171	777
Grad.Rate	118	78	65	53	10	17.17771	65.46332	777

Preprocessing the data

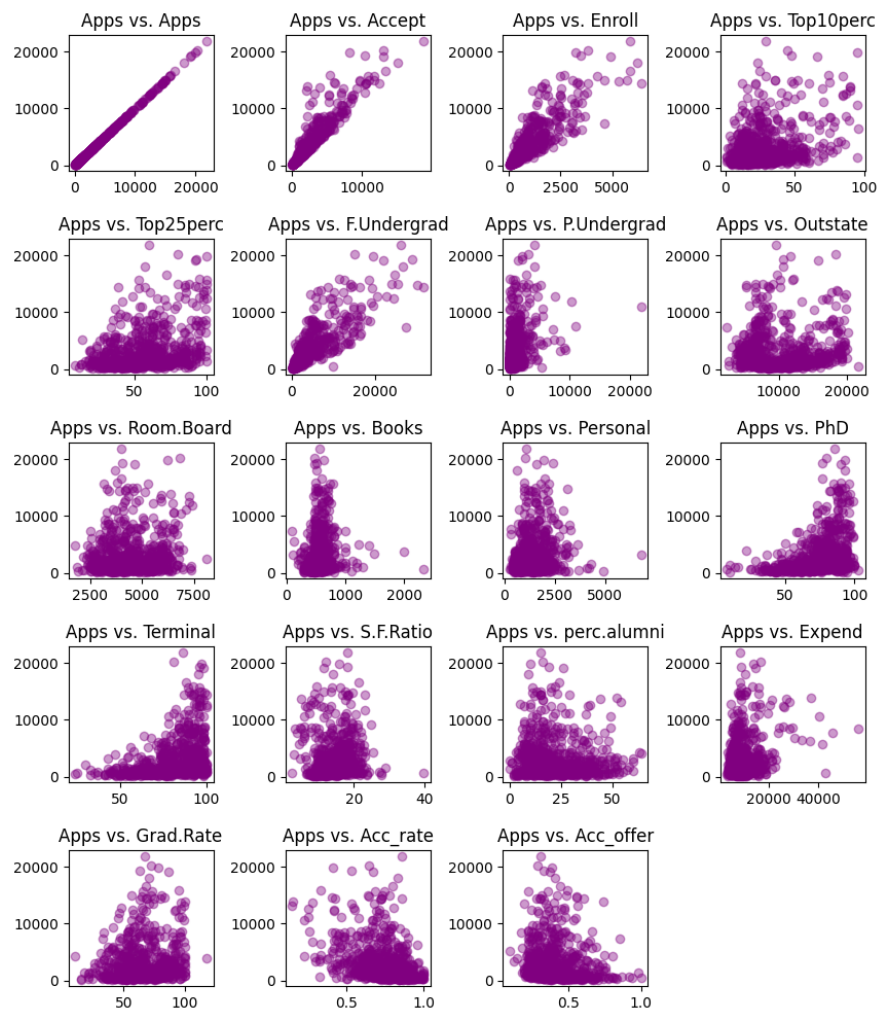
Preprocessing the data before modeling is surprisingly important because it has a significant effect on the quality of the results. Due to the quality of the dataset, there were no missing values or any duplicated rows. For data preparation, in the first step, we would change the categorical variable into a dummy variable, so it could be supported by machine learning models.

1. Is there any need to remove outliers?

As we discussed, because of the quality of the data, we didn't eliminate many outliers, using the famous Tukey method. However, we removed only one observation, Rutgers, and only because its response variable was more than twice as large as the next highest value.

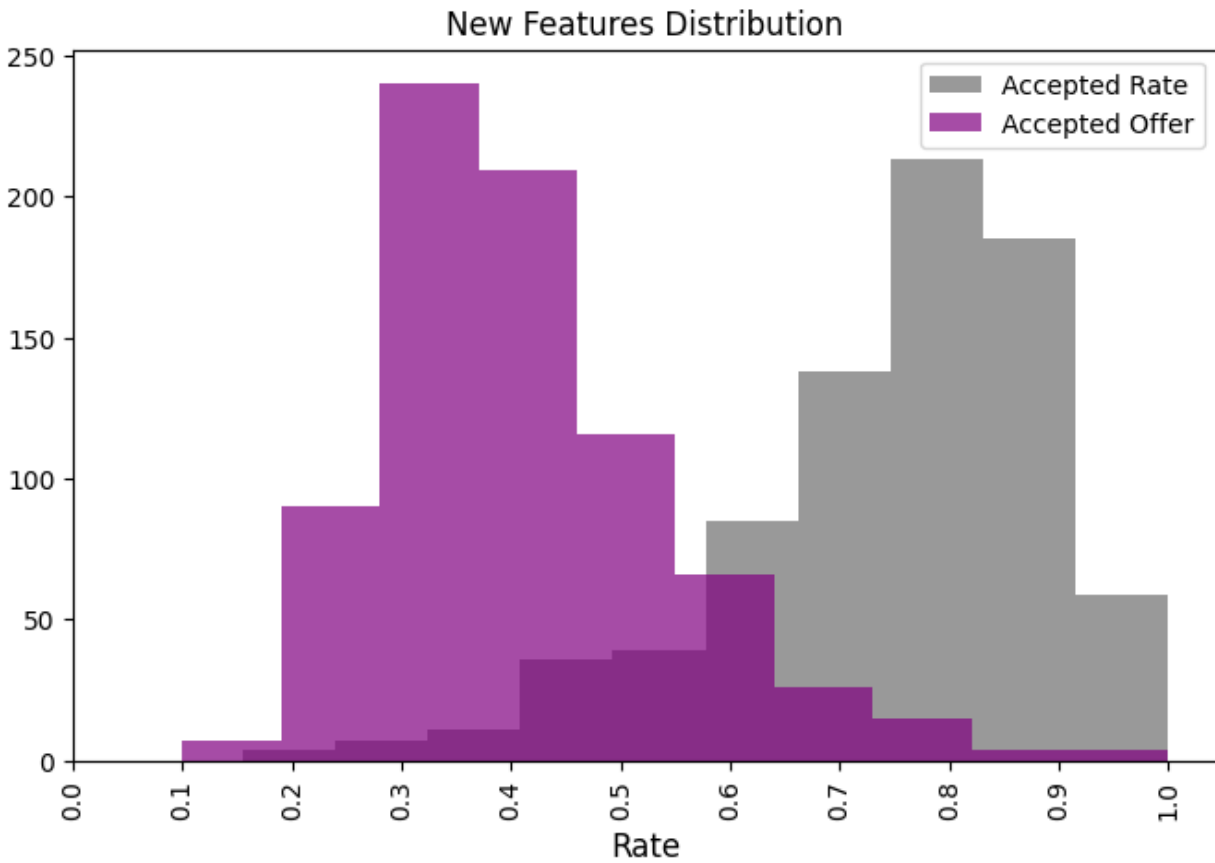
Also in the preprocessing, the names of colleges were removed, due to the fact that they don't reveal any insightful information.

These scatter plots illustrate the relationship among features and the response variable:



2. Is there any need for mixing the data?

For the feature engineering, two features were built, both of which were computed from other features. These features are as below:



The acceptance rate shows the ratio of people getting accepted in that college, and the accepted offer indicates the proportion of students who attended after they got accepted.

3. Is there any need to convert the data?

The dataset contains different features with different ranges of values. This can create a convergence problem while we want to optimize different models, especially deep learning models and neural networks. Hence, the data have to be scaled. In this project, we used a standard method for scaling independent variables. This standardization was held before using some of the models.

Furthermore, just before modeling, we have to split the data into train and test. In this project, due to the enough number of observations, we split the data with a proportion of 30 and 70 into the train and test

4. I there any need for a reduction in features?

Feature selection was unnecessary for this dataset, primarily due to its relatively small number of input variables and the absence of multicollinearity or redundant features. Preliminary univariate and bivariate data analysis showed that each feature contributed distinct and relevant information to the prediction task. Additionally, the model performance did not improve significantly after testing standard selection techniques such as correlation-based filtering or recursive elimination, suggesting that all variables held meaningful predictive value. Therefore, we retained all features to preserve the interpretability and integrity of the original data. Although there are some results comparing the performance of selected features, and without feature selection in the comparison metrics data frame.

Modeling and hyperparameter tuning

In this phase, various supervised learning algorithms were implemented to predict the target variable based on the input features.

Modeling is the core step in any machine learning workflow, as it enables the system to learn from historical data and make predictions on unseen cases.

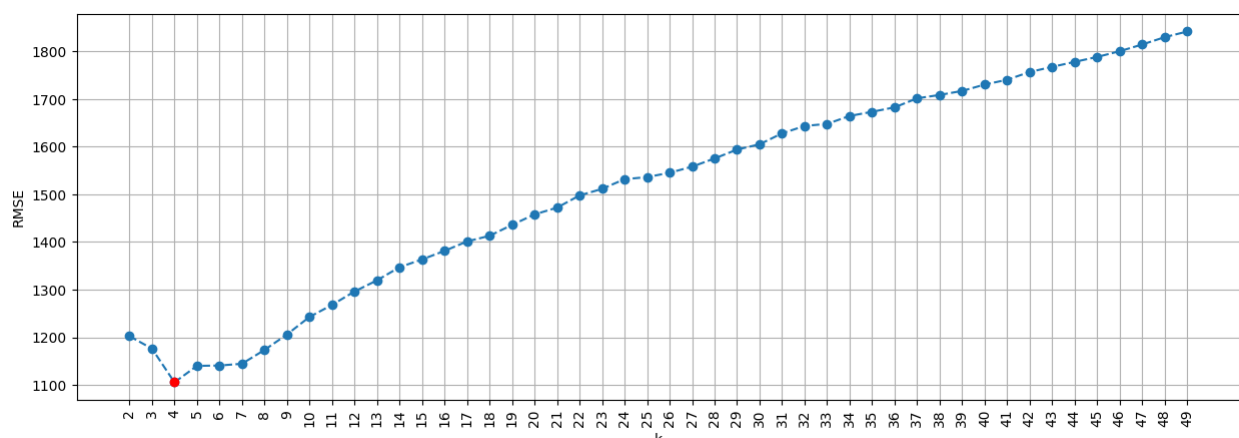
1. Building predictive models

In this project, a variety of different models, such as KNN, Linear Regression, Tree-based models, Gradient Boosting, and Neural Networks, were built to predict the response variable. All models have done a great job in predicting with considerable metrics. Among the various models, Neural Networks (DNN and RNN) did a decent job, and they had the best metrics among the others, which is nothing unusual since they are strongly capable of modeling non-linear relationships between dependent and independent variables.

The table comparing the models will be explained in more detail in the following paragraphs.

2. Model's hyperparameter tuning

Most of the models used in this project, such as Decision Tree, Random Forest, and K-Nearest Neighbors, require careful selection of hyperparameters to achieve optimal performance. To ensure robust evaluation and avoid overfitting, K-Fold cross-validation was employed, allowing each model to be trained and validated across multiple splits of the data. While hyperparameter tuning did not lead to significant performance improvements in all cases, selected models were fine-tuned using techniques such as Grid Search, and the tuned configurations are documented in the source code. The graph below is a sample of the optimal hyperparameter for and K-Nearest Neighbors model:



Evaluation of the built models

Each model was evaluated based on its ability to make accurate and consistent predictions on unseen data. Alongside standard regression metrics, qualitative analysis of model behavior helped identify overfitting or instability. The evaluation process played a key role in selecting the final model by balancing predictive performance, interpretability, and computational efficiency.

1. Validation of the model on the test data

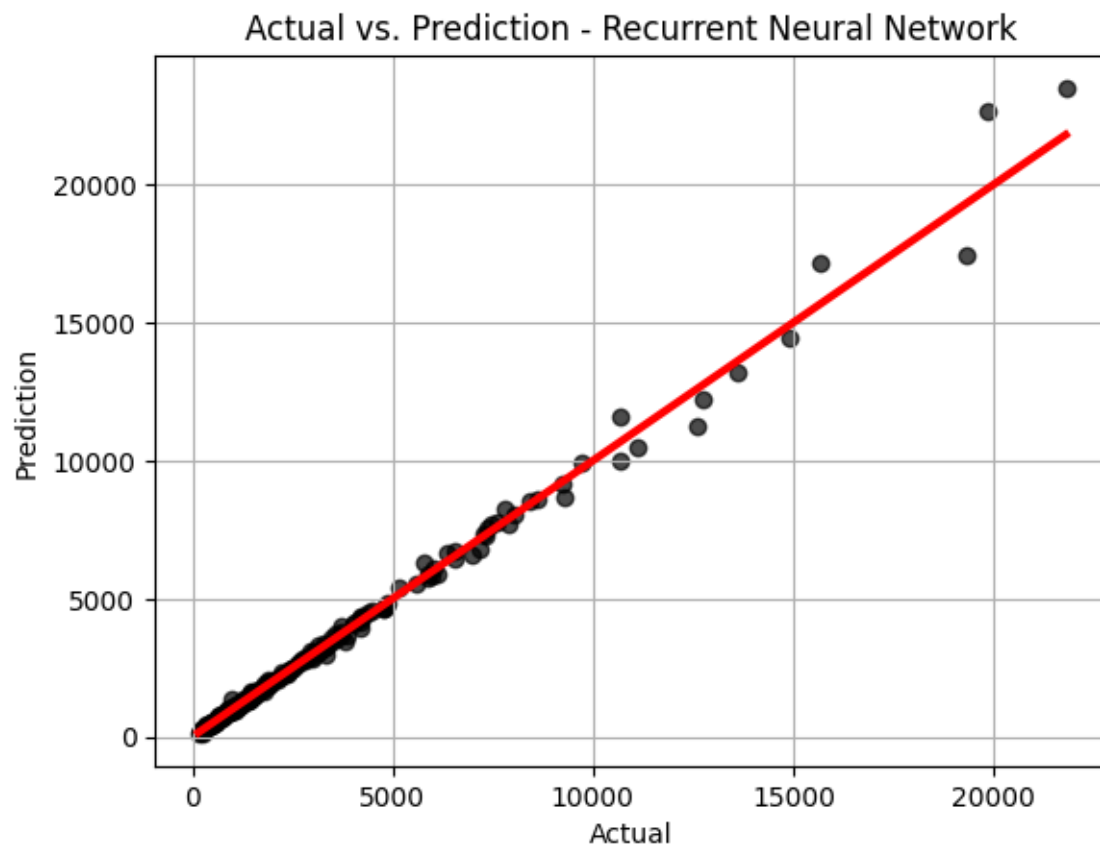
After training each machine learning model, a comprehensive evaluation was conducted using multiple performance metrics tailored to the regression nature of the task. Primary metrics included Mean Absolute Percentage Error, Root Mean Squared Error, R^2 score, and some others, which collectively assess both the accuracy and explanatory power of the predictions. These metrics were calculated on the test set to ensure that the results reflected the model's ability to generalize to unseen data. Visual tools such as residual plots and predicted vs. actual value charts were also used to support the quantitative findings.

The table below shows the model's performance in detail:

	MAPE	RMSE	MAE	Max of AbsErrors	Min of AbsErrors	IQR of AbsErrors	SD of AbsErrors	Median of AbsErrors
LR - All Features	39.261762	825.887321	491.538524	5333.132309	3.748909	433.924453	665.115302	304.811887
Knn- All Features	33.728947	1268.72983	703.479614	7425	0.25	706.25	1058.110089	340.25
Dtree - All Features	9.394953	912.075078	341.085386	6751	1.25	239.4	847.718077	94.6
RF - All Features	6.029889	830.908396	262.105961	7054.820275	0.836971	158.228762	790.182896	47.288579
SGB - All Features	5.28632	597.837796	180.66242	5206.418793	0.476768	116.033828	571.11382	43.368363
EGB - All Features	5.602449	701.11274	228.913981	6359.246094	0.405609	135.812378	664.116242	46.596588
DNN - All Features	20.06581	1173.30574	557.104613	7295.335938	2.372345	479.662048	1034.831798	216.857422
DNN - Scaled - All Features	4.570077	317.937911	123.039273	2815.541016	0.281494	79.906586	293.796372	49.494629
RNN - Scaled - All Features	4.430744	301.032729	118.705833	2544.933594	1.512024	76.418396	277.235456	41.84082
DNN - Scaled - Selected Features	7.416128	467.526916	167.171421	4680.347656	0.336304	96.481201	437.55781	27.386108
DNN - Scaled - Selected Features	8.52749	316.186981	172.230808	1998.884766	0.737793	88.821442	265.732608	98.59082
DNN - Scaled - Selected Features	12.189986	460.541401	255.882922	2393.630859	0.104492	206.099121	383.737287	123.077209
EGB - pca	5.310777	678.651569	224.496656	5888.430664	0.330139	138.68718	641.82332	44.824951

Among all the models evaluated, the Recurrent Neural Network (RNN) delivered the best predictive performance, outperforming both classical and other deep learning approaches. Although feature selection techniques were applied to improve generalization and reduce model complexity, they did not lead to any notable performance improvement. Likewise, applying Principal Component Analysis (PCA) had little effect, likely due to the relatively small number of input features and the reduction in interpretability caused by transforming the feature space. These results suggest that retaining the full set of original features and employing RNN-based architectures was the most effective strategy for this prediction task.

The graph below is the actual test dataset compared to the predicted data using the RNN model:



2. What suggestions do you have?

After seeing the results, two suggestions come to mind. First and foremost, in data science, better results always rely on better data, so any improvement in the quality of the data would make the results improve dramatically. Also, another thing that requires consideration is the data cleaning and outlier detection. As we discussed before, due to the quality of the dataset, none of the data

was eliminated. Nevertheless, most of the noise in the prediction is because of the observations with numerous applications. Hence, cleaning the outliers can make huge progress in the performance.

Second, collecting new features that can reveal information about the colleges can be beneficial.

Implementing the models

In this phase, the selected models, along with the necessary preprocessing steps, can be deployed into a web-based interface or application, where users are able to interact with the system by inputting new data and adjusting model hyperparameters. The platform enables users to experiment with various settings and immediately observe the impact of those changes on model performance.

1. What challenges could developing an algorithm have in the size of an industry?

Depending on the situation, developing predictive algorithms can have a variety of different challenges. For instance, some of the data which was collected and then used in this project can be confidential in the real world. Not many institutions would share data related to their performance, or even if they do, some may hesitate, so the predictions will be useless.

2. What suggestions do you have regarding these challenges?

In any data science project, before creating models, we have to think about how we can gather high-quality, reliable datasets. Anyone who wants to do a project related to prediction or forecasting has to take this initial step, because preparing data for prediction is as important as building machine learning models.

3. What requirement does your solution contain?

The main requirement that we have when we want to do this project in the real world is the full cooperation of the targeted colleges. Before doing anything, we have to make sure that we are able to collect enough data.

4. Develop a web application for your project

After predicting the target variable and evaluating the models, a web application was built. This web application allows people to change the hyperparameters of a K-Nearest Neighbors model to see the results.

This web application is designed as an interactive dashboard to facilitate exploratory analysis and evaluation of various classical machine learning regression models applied to a given dataset. Built

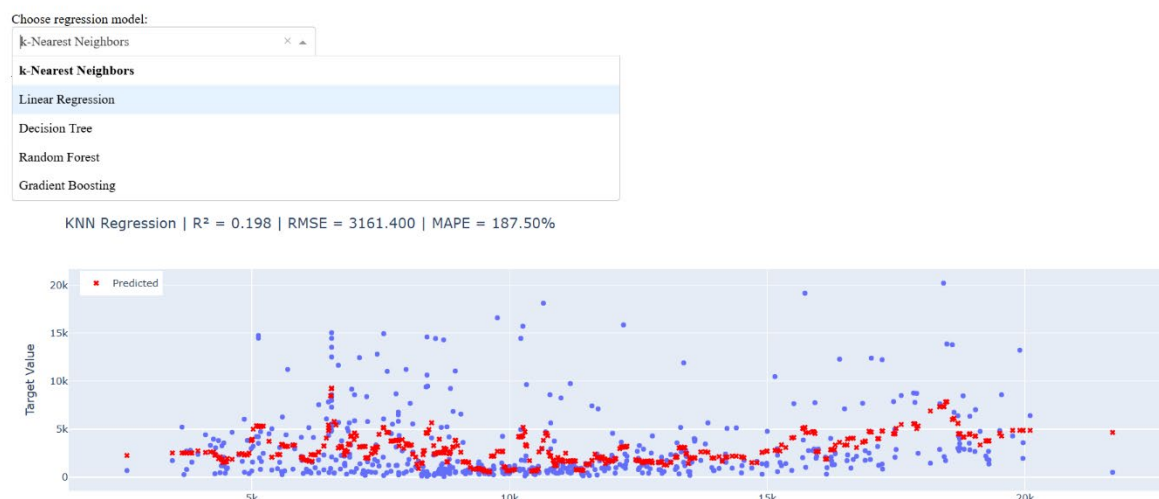
using the Dash framework, it provides users with an intuitive interface to select among multiple regression algorithms, such as k-Nearest Neighbors, Linear Regression, Decision Trees, Random Forests, and Gradient Boosting, and adjust their respective hyperparameters in real-time. This design empowers users to understand the effect of different models and tuning parameters on prediction performance without requiring deep programming knowledge.

The dashboard supports dynamic feature selection, allowing users to choose any numerical feature from the training dataset to visualize the relationship between that feature and the target variable. The selected feature serves as the explanatory variable on the x-axis in the scatter plot, while the actual and predicted target values are displayed on the y-axis. This flexible visualization capability aids users in exploring how individual features contribute to the predictive accuracy of each model, fostering a better grasp of feature importance and model behavior.

To quantitatively assess model performance, the dashboard calculates and displays several key regression metrics within the plot's title. These metrics provide immediate feedback on the error characteristics of the model predictions, enabling users to compare models systematically and select the most appropriate approach for their specific data and task.

All hyperparameter inputs for the supported models are incorporated within the dashboard interface but are conditionally shown or hidden based on the selected model. This design ensures a clean and user-friendly experience, preventing clutter and confusion from irrelevant inputs. The hyperparameters include, for example, the number of neighbors for K-Nearest Neighbors, maximum tree depth for decision trees, and number of estimators for ensemble methods and reflecting a practical balance between model flexibility and usability.

ML Regression Models Dashboard



Conclusion

1. What did you learn from doing this project?

One of the key lessons from this project was the undeniable importance of data quality in any data science or machine learning task. Through implementing different models and observing their outputs, it became clear that well-preprocessed data with informative features plays a crucial role in obtaining reliable predictions. Even the most sophisticated algorithms may perform poorly if the input data is messy, irrelevant, or lacks variance. As a result, I focused carefully on preparing the dataset by validating the values and ensuring the features chosen made sense about the target variable. In particular, the ability to dynamically explore the influence of each feature on the model performance, using the dashboard, further emphasized how important thoughtful feature selection and exploration are in predictive modeling. Moreover, I believe outlier observations can drastically influence the accuracy and reliability of models, specifically Linear Regression.

2. What challenges did you face and how did you solve them?

During the project, one of the main challenges was dealing with the relatively low prediction accuracy of simpler models such as Linear Regression and k-Nearest Neighbors. These algorithms, though interpretable and easy to implement, could not capture more complex, non-linear relationships in the data. As a solution, I moved toward more advanced machine learning algorithms, specifically, ensemble models like Random Forest and Gradient Boosting. These models significantly improved the regression metrics such as RMSE, R^2 , and MAPE. In another part of the project, I also experimented with Deep Neural Networks and Recurrent Neural Networks, which further enhanced the predictions by learning more abstract patterns within the data. These solutions demonstrated the value of using more complex models when faced with performance limitations from basic techniques.